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Nazarian; Benjamin et al.

Portable temperature controlled device

Abstract

A portable temperature-controlled device includes a housing and a controllable temperature element having a first surface and a second surface, wherein the first surface opposes the second surface. Inside the housing, a heat sink is disposed and in contact with the first surface of the controllable temperature element, and a fan is disposed adjacent the heat sink inside the housing and configured to direct heat away from the heat sink. The portable temperature-controlled device further includes a heat spreader in contact with the controllable temperature element for thermal energy transfer. The heat sink and the fan are supported on a support member inside the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 18/362,349, filed Jul. 31, 2023, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a temperature-controlled device, more specifically, a hand-held portable temperature-controlled device.

BACKGROUND

(2) Cooling and heating devices are used for therapeutic purposes or during surgical procedures due to many known benefits of hot and/or cold therapy in treatment. Accordingly, various apparatuses have been devised to achieve the desired heat and/or cold transfer. One issue with current heating and/or cooling devices is lack of portability and maneuverability. Bulky devices are often used to heat and cool the affected areas to provide fast and efficient heat transfer. The size of these devices is typically necessary because components are not optimally arranged or configured to promote efficient operation. These heating and cooling devices are often heavy, need to be in a fixed position, and plugged into a continuous power supply. Smaller and lighter multi-therapy devices are desirable. One potential hurdle to a smaller form factor has been managing heat dissipation within a portable device to protect the internal components from overheating. Accordingly, a need exists for a light portable device that provides heating and cooling therapy comparable to the bulky device while also providing efficient heat management. It is further desirable to provide a versatile multi-therapy device that can be used on different body parts of a user, whether manually by the user or, if desired, in a fixed position.

(3) The background description disclosed anywhere in this patent application includes information that may be useful in understanding the present disclosure. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed concepts, or that any publication specifically or implicitly referenced is prior art.

SUMMARY

(4) Described herein is a novel portable temperature-controlled device for therapeutic applications including both cold and heat therapies within a compact and handheld device. In some aspects, the portable temperature-controlled device is assembled with an adjustable strap system to wrap around a patient body part. The temperature-controlled device is designed to have an ergonomic configuration to be comfortably held by one hand, while placing the device on a desired area of the body. These areas can include the back, knee, elbow, shoulder, ankle, etc. The temperature-controlled device can be further assembled with a strap to secure on the body part to free the user's hand.

(5) The temperature-controlled device can include a thermoelectric element to transfer heating and cooling to the user's body. In a typical implementation, the thermoelectric element is a device that utilizes the Peltier Effect, having one side of which cools while the other side heats. The heating and cooling modes can be optionally and selectively switched within predetermined temperature ranges. The temperature-controlled device can further include a rechargeable battery for prolonged use.

(6) In one aspect, a hand-held portable temperature-controlled device includes a housing having a generally cuboid shape with curved surfaces and configured to be grasped by one hand. The housing may include: a controllable temperature element having a first surface and a second surface and configured to generate cooling and heating; a heat sink disposed on the first surface of the controllable temperature element; a fan disposed on the heat sink and configured to direct heat away from the heat sink; a heat spreader comprising a first side and a second side, the first side extending out from the housing and contacting a user's body part; a support member configured to support the heat sink and the fan; and a temperature controller connected to the controllable temperature element. The housing further includes a first air inlet configured to permit air flow into the housing and an air outlet configured to permit air flow to flow out of the housing, and the first air inlet and the air outlet are in fluid communication with each other.

(7) In another aspect, a wearable assembly includes a temperature-controlled device which comprises a housing having a generally cuboid shape. The housing may include: a controllable temperature element having a first surface and a second surface; a heat sink disposed on the first surface of the controllable temperature element; a fan disposed on the heat sink and configured to direct heat away from the heat sink; a heat spreader comprising one side and another side and configured to receive thermal energy from the controllable temperature element; and a

support member configured to support at least one battery, the heat sink, and the fan. The wearable assembly includes a strap case configured to be assembled to the temperature-controlled device. The strap case may comprise: a center opening through which the heat spreader extends on a bottom side; and a pair of side arms curvedly, integrally extending upward, each having a protrusion protruding toward the center opening.

(8) Further features and advantages, as well as the structure and operation of various aspects, are described in detail below with reference to the accompanying drawings. It is noted that the specific aspects described herein are not intended to be limiting. Such aspects are presented herein for illustrative purposes only. Additional aspects will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate aspects of the present disclosure and, together with the description, further serve to explain the principles of the disclosure and to enable a person skilled in the pertinent art to make and use the disclosure.
- (2) FIG. 1 is a perspective view of a portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (3) FIG. 2 is another perspective view of the portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (4) FIG. 3 is a side view of the portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (5) FIG. 4 is a top view of the portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (6) FIG. 5 is a bottom view of the portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (7) FIG. 6 is an exploded view of the portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (8) FIGS. 7 and 8 are cross-sectional views of the portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (9) FIG. 9 is an interior of a lower housing of the portable therapeutic temperature-controlled device according to aspects of the present disclosure.
- (10) FIG. 10 is an exploded view of FIG. 9.
- (11) FIG. 11 is a top view of the portable therapeutic temperature-controlled device grasped by a user's hand.
- (12) FIG. 12 is a side view of the portable therapeutic temperature-controlled device grasped by a user's hand.
- (13) FIG. 13 is a perspective view of a strap case according to aspects of the present disclosure.
- (14) FIG. 14 is a perspective view of the strap case of FIG. 13 and the portable therapeutic temperature-controlled device of FIG. 1.
- (15) FIG. 15 is a perspective view of the strap case and the portable therapeutic temperature-controlled device as assembled.
- (16) FIG. 16 is a side view of the strap case and the portable therapeutic temperature-controlled device as assembled.
- (17) FIG. 17 is a top view of the strap case and the portable therapeutic temperature-controlled device as assembled.
- (18) FIG. 18 is a bottom view of the strap case and the portable therapeutic temperature-controlled device as assembled.
- (19) FIG. 19 illustrates a magnified cross-section view of FIG. 16.
- (20) FIGS. 20A and 20B are top and bottom views of a primary strap according to aspects of the present disclosure.
- (21) FIGS. 21A and 21B are top and side views of the portable therapeutic temperature-controlled device assembled with the primary strap.
- (22) FIGS. 22A and 22B are top and bottom views of a secondary strap according to aspects of the present disclosure.
- (23) FIGS. 23A and 23B are top and side views of the portable therapeutic temperature-controlled device assembled with the primary strap and the secondary strap.
- (24) In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

(25) The following descriptions and drawings are illustrative and are not to be construed as limiting. Numerous specific details are described to provide a thorough understanding of the disclosure. However, in certain instances, well-known or conventional details are not described in order to avoid obscuring the description. References to one or more aspects in the present disclosure can be, but not necessarily are references to the same aspect; and, such references mean at least one of the aspects. If a component is not shown in a drawing then this provides support for a negative limitation in the claims stating that that component is “not” present. However, the above statement is not limiting and in another aspect, the missing component can be included in a claimed aspect.

(26) Reference in this specification to “one aspect,” “an aspect,” “a preferred aspect” or any other phrase mentioning the word “aspect” means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect of the disclosure and also means that any particular feature, structure, or characteristic described in connection with one aspect can be included in any aspect or can be omitted or excluded from any aspect. The appearances of the phrase “in one aspect” in various places in the specification are not necessarily all referring to the same aspect, nor are separate or alternative aspects mutually exclusive of other aspects. Moreover, various features are described which may be exhibited by some aspects and not by others and may be omitted from any aspect. Furthermore, any particular feature, structure, or characteristic described herein may be optional. Similarly, various requirements are described which may be requirements for some aspects but not other aspects. Where appropriate any of the features discussed herein in relation to one aspect or aspect of the disclosure may be applied to another aspect or aspect of the disclosure. Similarly, where appropriate any of the features discussed herein in relation to one aspect or aspect of the disclosure may be optional with respect to and/or omitted from that aspect or aspect of the disclosure or any other aspect or aspect of the disclosure discussed or disclosed herein.

(27) The terms used in this specification generally have their ordinary meanings in the art, within the context of the disclosure, and in the specific context where each term is used. Certain terms that are used to describe the disclosure are discussed below, or elsewhere in the specification, to provide additional guidance to the practitioner regarding the description of the disclosure. For convenience, certain terms may be highlighted, for example using italics and/or quotation marks. The use of highlighting has no influence on the scope and meaning of a term; the scope and meaning of a term is the same, in the same context, whether or not it is highlighted.

(28) It will be appreciated that the same thing can be said in more than one way. Consequently, alternative language and synonyms may be used for any one or more of the terms discussed herein. No special significance is to be placed upon whether or not a term is elaborated or discussed herein. Synonyms for certain terms are provided. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms discussed herein is illustrative only, and is not intended to further limit the scope and meaning of the disclosure or of any exemplified term. Likewise, the disclosure is not limited to various aspects given in this specification.

(29) Without intent to further limit the scope of the disclosure, examples of instruments, apparatus, methods and their related results according to the aspects of the present disclosure are given below. Note that titles or subtitles may be used in the examples for convenience of a reader, which in no way should limit the scope of the disclosure. Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure pertains. In the case of conflict, the present document, including definitions, will control.

(30) It will be appreciated that terms such as “front,” “back,” “top,” “bottom,” “side,” “short,” “long,” “up,” “down,” “aft,” “forward,” “inboard,” “outboard” and “below” used herein are merely for case of description and refer to the orientation of the components as shown in the figures. It should be understood that any orientation of the components described herein is within the scope of the present disclosure.

(31) The terms “connected” or “coupled” and related terms are used in an operational sense and are not necessarily limited to a direct connection or coupling. The term “thermally coupled” means coupled in a way capable of conducting heat, and the term “thermally insulated” means separated by a substance that deters heat transfer.

(32) The term “flexible” generally means bendable and adaptable under relatively little force. In the context of various aspects of the present disclosure, flexible is intended to describe the dynamic conforming nature of the personal temperature controlled device to the general shape of a portion of a person's body, such as wrist, ankle, neck, shoulder, back, chest, forehead, rib cage, arch, temple, palm, etc., directly or indirectly in contact with or otherwise engaging a surface of the personal temperature-controlled device. In addition, the term “approximately” is generally used to modify a numerical value above and below the set value by a variation of $\pm 10\%$.

(33) Described herein and shown in FIGS. **1-10** is an exemplary temperature-controlled device that can provide cooling and heating effects. FIGS. **11** and **12** show the temperature-controlled device grasped by a user's hand.

(34) FIG. **1** shows a temperature-controlled device **10** designed for a user to comfortably grasp by one hand while providing a heating or cooling therapeutic effect to a desired area on the user's body and also promoting air flow into and out of the temperature-controlled device **10**. In some aspects, the temperature-controlled device **10**

includes a housing **100**, on an outer side of which includes an air inlet or a first air inlet **102** on a top surface of the temperature-controlled device **10**, an air outlet **104** on at least one side surface (e.g., at least one of a first side surface or a second side surface of the temperature-controlled device **10**, a pair of recesses **106** on opposing side surfaces, a plurality of buttons **108** for selectively controlling a heating mode and a cooling mode of a temperature mode by user manipulation, a first or control mode Light-Emitting Diode (LED) **110** disposed below the buttons **108** for indicating the temperature mode, a Universal Serial Bus (USB) charging port or simply charging port **112**, a second or battery level LED **114**, and an undercut **116**. In some aspects, the first LED **110** may include one or more LEDs. In some aspects, the overall shape of the housing **100** may be curvedly shaped and dimensioned for ergonomic holding by a user's hand. In further aspects, each element of the temperature-controlled device **10** is arranged and designed to provide efficient operation of the heating and cooling effect and optimal heat dissipation in a compact space, as will be described later. For example, the housing **100** may be rectangular in shape to allow for one-handed holding while maximizing the size of air inlet **102**.

(35) Referring to FIGS. **2** and **3**, in some aspects, the temperature-controlled device **10** further includes a heat spreader **118** on the bottom side of the housing **100**. In some aspects, the heat spreader **118** may be implemented as a metallic heat spreader and can be made of high thermal conductive materials, such as, for example, copper, aluminum, or other metals or metal alloys, or certain ceramics, to help transfer heat or cold to increase the effective area of heating or cooling treatment. As will be described later, the housing **100** has an opening (e.g., “**146**” in FIGS. **6** and **10**) defined in the bottom side of the housing **100** through which the heat spreader **118** extends out of the housing **100**. While the housing **100** may be formed of, e.g., a plastic material, the bottom portion of the housing **100** exposes the heat spreader **118**, which in some aspects may be constructed of a metal material. Thus, the heat spreader **118** may have one side exposed outside that can directly transfer thermal energy to a body part of the user. In addition, this configuration allows engagement with a strap case, which will be described later, while ensuring that the exposed side of the heat spreader **118** maintains direct contact with the user's skin.

(36) Referring to FIGS. **3** and **4**, in some aspects, the temperature-controlled device **10** has a generally cuboid shape with curved surfaces and may have a dimension that a user can easily grasp by one hand. For example, the housing **100** may have a length approximately between 110 mm and 115 mm (L), a width approximately between 75 mm and 80 mm (W), and a height approximately between 52 mm and 58 mm (H). This is one exemplary dimension such that each of length, width, and height may be smaller or greater than what's described herein. The housing **100** may be formed of a plastic material, e.g., one or more bonded layers of one or more of polyester (polyether), polyethylene, polypropylene, nylon, kevlar, nomex, polyacrylonitrile, cellulose, polyurethane, polycarbonate, and acrylonitrile-butadiene-styrene terpolymer, or similar foams and/or fibers. However, it is not limited to the listed materials for the housing **100**, but any non-metallic materials, metallic, wood, or the like could be properly implemented. In addition, the housing **100** is curvedly shaped so as to be ergonomic and to facilitate gripping by a user's hand.

(37) To promote proper one-handed gripping, the upper portion of the outer side surfaces has grip portions **148** inwardly curved forming a concave surface to provide comfort grasp particularly for fingers. Such a shape can induce a user to hold the temperature-controlled device **10** in a certain position as shown in FIGS. **11** and **12**. The shape and dimensions of housing **100** are configured so as to prevent the buttons **108**, the first LEDs **110**, the charging port **112**, and the second LED **114** from being accidentally gripped or covered when the user is holding the temperature-controlled device **10**. Further, the size and dimensions of the air inlet **102** and the air outlet **104** are configured such that they are exposed, partially or entirely, when a user is gripping the temperature-controlled device **10**, which facilitates air flow within the housing **100** even when the temperature-controlled device **10** is being held. However, the shape is not limited to what is shown in the drawings or described above. In other words, the temperature-controlled device **10** may have a cylindrical shape, a spherical shape, a hemispheric shape, a prism shape (e.g., triangular prism, hexagonal prism, pentagonal prism, or the like), a cone shape, a tetrahedron shape, an octahedron shape, etc.

(38) The configurations according to various aspects enable the compact size and light weight of the temperature-controlled device **10**, as well as efficient and portable power management, which enables a user to engage in an active life-style while still obtaining a thermal therapy. Further, overheating of the temperature-controlled device **10** can be prevented.

(39) Referring back to FIG. **3**, the housing **100** may include individually configured parts assembled together, including a top cover **120** having a convex shape, an upper side cover **122**, and a lower side cover **124**. The top cover **120** forms a top portion of the housing **100**, the upper side cover **122** is assembled to the top cover **120**, and the lower side cover **124** is then assembled to the upper side cover **122**. Each cover may be configured with a curved surface such that the top cover **120**, the upper side cover **122**, and the lower side cover **124** together form a convex contour of the housing **100** for ergonomic design. In addition, the upper side cover **122** and the lower side cover **124** are assembled forming a partially enclosed space of the housing **100** together with the top cover **120**.

(40) The top cover **120** may include a first plurality of vents or openings that form the air inlet **102** to be formed on

an upper surface or top surface of the housing **100**. The first plurality of vents may be defined on the entire surface, or partially, of the top cover **120**. The air inlet **102** may be configured to guide air flow into the housing **100** through the first plurality of vents. For instance, a fan (“**126**” in FIG. **4**) may be located adjacently under the top cover **120** to cause the air flow to flow into the housing **100** through the first plurality of vents. The lower side cover **124** may be formed with a second plurality of vents or openings on the longer (longitudinal) side of the lower side cover **124** of the housing **100** as the air outlet **104** to permit the air flow to flow or exit out from the housing **100**, on the shorter (width) side of the lower side cover **124**, or a combination of both. Accordingly, the first plurality of vents and the second plurality of vents are configured to define an air flow passage for the air flow in and out of the housing **100**. In addition, an air flow passage may be formed from the first plurality of vents as the air inlet **102** and the undercut **116** as a third opening to the second plurality of openings as the air outlet **104** through the fan **126** and the heat sink **132**.

(41) Alternatively or additionally, the second plurality of vents or openings may be formed on one side surface, two side surfaces (e.g., opposite side surfaces), three side surfaces, or all four side surfaces of the lower side cover **124**. For instance, FIG. **3** shows the air outlet **104** as the second plurality of vents or openings formed on a first surface, while FIGS. **9** and **10** show the air outlet **104** formed on the first surface and a second surface which faces the first surface. In some aspects, the second plurality of vents are arranged longitudinally along the corresponding surface. The plurality of vents or openings of the lower side cover **124** may form the air outlet **104**, which may be configured to facilitate the air flow out of the housing **100**. In some aspects, the air outlet **104** may be provided on opposite longer (longitudinal) sides or surfaces of the lower side cover **124** (e.g., the first and second side surfaces facing each other) to guide inside air to flow out of the housing **100** in opposing directions. In some aspects, the air outlet **104** is formed on side surfaces of the housing **100** so as to avoid air flowing out from the air outlet **104** does not blow onto a user's skin. In some aspects, the air outlet **104** is positioned on the side surface the housing **100** so as to prevent the user from being able to obstruct air flow while holding the housing **100**.

(42) The housing **100** may further include a fan **126** configured to draw or pull outside air into the housing **100**. Referring to FIG. **4** which shows a top view of the temperature-controlled device **10**, the fan **126** may be located directly, adjacently below the top cover **120** for drawing outside air into the housing **100** through the air inlet **102**. The fan **126** is configured to circulate the air inside the housing **100** and guide out, through the air outlet **104**, to exit the housing **100**.

(43) Referring back to FIG. **3**, the pair of recesses **106** that are surface-treated may be formed on opposing side surfaces (e.g., third and fourth surfaces positioned facing each other in a width direction) of the lower side cover **124**. In some aspects, the recesses **106** are formed on the sides where the air outlet **104** are not defined. For example, the air outlet **104** may be formed on the longer (longitudinal) side or sides of the lower side cover **124** while the recesses **106** may be formed on the shorter (width) side or sides of the lower side cover **124** as illustrated in FIG. **3**. In other aspects, the air outlet **104** and the recesses **106** may be configured on the same side of the lower side cover **124**. The recesses **106** may be configured to receive a strap case which will be described below.

(44) In some aspects, the upper side cover **122** may include an undercut **116** configured as a recessed surface positioned along an upper edge of the upper side cover **122** adjacent to the top cover **120**. The undercut **116** can form a gap between the top cover **120** and the upper side cover **122** and is configured to facilitate additional and/or alternative air to flow into the housing **100**. That is, in some aspects, the undercut **116** may be implemented with the air inlet **102** as a second air inlet such that air is permitted to flow into the housing **100** through the air inlet **102** and additional air through the undercut **116**, as shown by the arrows in FIG. **3**. In other aspects, the housing **100** may comprise only one of the air inlet **102** and the undercut **116** to receive air.

(45) When a user is grasping the temperature-controlled device **10** by one hand, with reference to FIGS. **11** and **12**, the air inlet **102** may be, partially or entirely, blocked and air flow into the housing **100** may be partially obstructed by the user's hand. In any situation, the undercut **116** is configured to provide an additional air passage for additional air to be drawn into the housing **100** to supplement the air flow of the air inlet **102**. The air inlet **102**, the undercut **116**, and the fan **126** are configured to facilitate air flow through a central path within an internal cavity of the housing **100** of the temperature-controlled device **10**. For instance, the air inlet **102** together with the undercut **116** and the air outlet **104** are in fluid communication with each other via the central path within the internal cavity (for example, see shaded area “A” in FIG. **7**). Maximal air flow in housing **100** can be provided by the combination of the air inlet **102**, the undercut **116**, and the fan **126**, along with the physical arrangement of components within housing **100** within the temperature-controlled device **10**. The increased air flow that results from this arrangement allows the temperature-controlled device **10** to operate efficiently (e.g., fast cooling, fast heating) in the heating, cooling, and contrast modes in a portable handheld device.

(46) Referring to FIG. **5**, the bottom portion of the housing **100** has an opening (e.g., “**146**” in FIGS. **6** and **10**), which will be described later, through which the heat spreader **118** extends out of the housing **100**. In some aspects, the heat spreader **118** is configured with dimensions to extend outside of the housing **100** and beyond a strap case **20** when it is attached to housing temperature-controlled device **10** (see FIG. **14**). The heat spreader **118** may be

dimensioned so that it is in contact with a skin surface of the user when the temperature-controlled device **10** is used with or without strap case **20**. The heat spreader **118** may have a convex-shaped bottom surface to facilitate contact with surface of the user's body and align with the overall curved shape of the housing **100**. With such a convex configuration, the amount of material used for the heat spreader **118** may be reduced while maintaining or improving the heat transfer performance. In addition, the heat spreader **118** may be configured to increase or decrease the temperature of about, e.g., 48° C. in about, e.g., 150 seconds, so as to quickly and efficiently deliver the desired temperature to a user. As will be further described later, the heat spreader **118** is designed for material/weight reduction to satisfy the compactness. In some aspects, the heat spreader **118** may be formed of a conductive alloy, metal, or material, such as aluminum, stainless steel, carbon-fiber, or carbon-carbon materials and/or composites. The dimensions of the spreader **118** may be tailored for the particular application. Such dimensions as well as the shape may be configured to reduce amount of material used while increasing the contact area with a user. For example, curved outer surfaces, rather than flat surfaces, can enhance the contact with the user's body, or unnecessary materials can be removed to reduce the overall size and weight.

(47) Internal components of the temperature-controlled device **10** and their arrangement within housing **100** will be described in detail with reference to FIGS. **6-10**.

(48) FIG. **6** is an exploded view of the temperature-controlled device **10**, where each case part of the housing **100** is opened to show internal components of the temperature-controlled device **10** enclosed inside the housing **100**. As described above, the temperature-controlled device **10** includes the fan **126** that may be composed of a fan blade portion **127** and a fan housing **128**. The fan **126** may be arranged adjacently below the top cover **120** inside the housing **100** to pull ambient air into the housing **100** via the air inlet **102**. In some aspects, a heat sink **132** may be arranged beneath the fan **126**. The heat sink **132** may be disposed on a first surface of controllable temperature element **134**, which can generate cooling and heating. In some aspects, the first surface of controllable temperature element **134** is an upper surface. Examples of the controllable temperature element **134** include a Peltier device, Peltier heat pump, solid state refrigerator, thermoelectric cooler (TEC), etc. The controllable temperature element **134** may adapt the Peltier effect to create a heat flux at the junction of two different types of materials, and transfer heat from one side of the device to the other. In some aspects, a second surface of the controllable temperature element **134** is in contact with a first side (or one side) of the heat spreader **118** so as to transfer thermal energy to the upper surface of the heat spreader **118**. In some aspects, the second surface may oppose the first surface; for example, when the first surface is implemented as an upper surface of controllable temperature element **134**, the second surface may be implemented as a lower surface of controllable temperature element **134**. A second side (or another side) of the heat spreader **118** that opposes the first side of the heat spreader **118** may be disposed inside the housing **100**. The first side of the heat spreader **118** may extend out from the housing **100**.

(49) The heat sink **132** can pull heat from the upper surface of the controllable temperature element **134**, and the fan **126** can help dissipate heat or direct heat away from the heat sink **132** and other components. When the temperature-controlled device **10** is in use, the heat spreader **118** is cooled or heated by the controllable temperature element **134** and the heat or cold can be transferred to, by contacting, a user via the heat spreader **118**. In some aspects, the temperature-controlled device **10** further includes a printed circuit board or PCB **140** for electrical and data communication. The PCB **140** may include a control unit (e.g., temperature controller) **142** connected to communicate with the controllable temperature element **134** for various control functions (e.g., turning on or off the temperature-controlled device **10**, heating or cooling, etc.).

(50) The controllable temperature element **134** may be implemented as a heat pump that can directly convert electricity into heating and cooling power depending on the mode of the temperature-controlled device **10**. When power is supplied to the controllable temperature element **134**, the current causes one side (cool side) of the controllable temperature element **134** to absorb heat. The opposite side of the controllable temperature element **134** may then release heat (the hot side). For instance, when a user presses one of buttons **108** for a heating mode, one side of the controllable temperature element **134** releases heat where the one side is in contact with the user. When a user presses one of buttons **108** for a cooling mode, on the other hand, the same side of the controllable temperature element **134** may absorb heat rather than releasing the heat to provide a cooling effect to the user. Further, during a contrast mode, the controllable temperature element **134** may be controlled to periodically alternate heat release and absorption functions. That is, the controllable temperature element **134** causes heat to flow from the cool side to the hot side. Reversing the current causes the heat to be moved in the opposite direction thereby reversing the hot side and the cold side. Consequently, the heating or cooling effect can be selectively performed. Based on the disclosure provided herein, one of ordinary skill in the art will recognize the various possible reconfigurations of temperature-controlled device **10** that would achieve a heating/cooling effect.

(51) The controllable temperature element **134** may have dimensions of approximately, e.g., 40 mm (L)×40 mm (W)×4.8 mm (D), while the heat sink **132** has a surface area slightly greater than the controllable temperature element **134**. The fan **126** may have the overall dimensions of approximately, e.g., 60 mm (L)×60 mm (W)×10 mm (D). The heat spreader **118** may have dimensions of approximately, e.g., 70 mm (L)×60 mm (W)×7 mm (D). The

disclosed dimensions or to be limited to these numeric values and may be configured within a range of sizes to maintain the ability for one-handed gripping of the temperature-controlled device **10**.

(52) The temperature-controlled device **10** further includes one or more batteries **130** which include at least one battery. In some aspects, the one or more batteries **130** may be one or more rechargeable batteries and may be arranged on one or more sides of the perimeter of the heat sink **132**. Referring to FIG. 7, which is a cross-sectional view of FIG. 1, the one or more batteries **130** may include two rechargeable batteries seated on an outer side of a support member **137**. The support member **137** may be located inside the housing **100**, horizontally off-centered, between the fan **126** and the controllable temperature element **134**. The support member **137** may have a square-shaped bottom side on which the heat sink **132** is seated at a middle portion of the support member **137**. Two side walls **138** of the support member **137** may extend straightly upward. Each of the two side walls **138** may include a wing portion **139** having a shape for supporting the one or more rechargeable batteries **130** as shown in FIGS. 6-8. In some aspects, one or more side walls of the support member **137** extend as described above to support the corresponding number of rechargeable battery. Accordingly, the support member **137** can support the one or more batteries **130**, the heat sink **132**, and the fan **126** in a manner that facilitates air flow between the fan **126** to heat spreader **118**. Support member **137** is configured to hold the one or more batteries **130** in a horizontally spaced position such that the one or more batteries **130** do not impede air flow within housing **100**. In an alternative embodiment, the one or more batteries **130** may be external to the housing **100** and connected to the temperature-controlled device **10** by, for example, a charging cable.

(53) Referring back to FIG. 6, a bottom side of the support member **137** has, at its center, an opening **144** through which the controllable temperature element **134** is fitted to be in contact with the heat sink **132** below the heat sink **132**. In addition, the heat spreader **118** may be connected to the bottom side of the support member **137**. In some aspects, the opening **144** may be smaller than the heat sink **132** but larger than the controllable temperature element **134**. In addition, a padded liner **136** (FIG. 10) having a non-conductive material may be implemented on the second side of the heat spreader **118** inside the housing **100** to protect the controllable temperature element **134** and its surroundings to prevent thermal energy from transferring to unnecessary areas within housing **100**, by surrounding the controllable temperature element **134**.

(54) The above-described configuration and structure of the internal components can provide a rapid heating and cooling effect to a user through direct contact with the heat spreader **118** while allowing for efficient cooling of the internal components of temperature-controlled device **10**. For instance, when a user grasps the temperature-controlled device **10** as shown in FIGS. 11 and 12 for hot or cold therapy, outside air can sufficiently enter the temperature-controlled device **10** through the combination of the air inlet **102** and the undercut **116** by operating the device, and the air can be guided to flow down to the fan **126**, the heat sink **132**, and finally to outside the housing **100** through the air outlet **104**, enabling fast cooling.

(55) In some aspects, the operation (e.g., heating and cooling) of the temperature-controlled device **10** can be controlled by user manipulation of the buttons **108**. Referring to FIGS. 7 and 8, the temperature-controlled device **10** may include two or more buttons **108** connected on the PCB **140**, where the buttons **108** are on the external surface of the housing **100** for a user to press. The buttons **108** can control turning on and off of the temperature-controlled device **10**, changing the control modes, changing temperature settings, etc. Some of the features are controlled by multiple presses of the associated button. In one aspect, one of the buttons **108** may control a heating mode, and the other one of the buttons **108** may control a cooling mode.

(56) More specifically, the buttons **108** may include one button for turning on the temperature-controlled device **10**, and the first LEDs **110** may display, e.g., green light, to indicate that the temperature-controlled device **10** is on. The buttons **108** may also include corresponding buttons for activating different modes of the temperature-controlled device **10**, such as a button configured to activate (and deactivate) a heating mode and a button configured to activate (and deactivate) a cooling mode. The buttons **108** may be visually distinguishable from each other by icons or colors, e.g., red for heating mode and blue for cooling mode. In addition, two or more buttons of buttons **108** may be configured to be pressed simultaneously to activate a contrast mode. In some aspects, the temperature-controlled device **10** may be configured to detect the length of button presses for the buttons **108**. For example, pressing a button for a predetermined period of time (e.g., 2 seconds or less) may cause the temperature-controlled device **10** to switch modes. In some aspects, the predetermined period of time may be different (e.g., shorter or longer) than the period of time needed to press the button for activating a mode or turning on the temperature-controlled device **10**.

(57) The heating and cooling modes may be configured with different temperature levels. The heating mode may implement a range of temperature values including predetermined heating thresholds (e.g., above 30° C. but less than 50° C., or alternatively between 35° C. and 43° C.) as a first temperature setting. Examples of discrete values within the range for the heating mode may include, for example, values of 35° C., 39° C., and 43° C. In some other aspects, the temperature range for each mode may vary, such that each range may be greater (e.g., above 20° C. but less than 60° C.) or smaller (e.g., above 38° C. but less than 40° C.).

(58) Similarly, the cooling mode may implement a range of temperature values including a predetermined cooling threshold (e.g., below 20° C., or alternatively between 8° C. and 16° C.) as a second temperature setting. Examples of discrete values within the range for the cooling mode may include, for example, values of 16° C., 12° C., and 8° C. In some aspects, the heating and cooling temperature values of the temperature-controlled device **10** may be specifically chosen to maximize the benefits and safety of treatment directly on users' skin without concern for skin irritation, burns, etc. In some other aspects, the temperature range for each mode may vary, such that each range may be greater (e.g., above 3° C. but less than 30° C.) or smaller (e.g., above 10° C. but less than 18° C.).

(59) The buttons **108** may include one or more buttons for configuring the temperature settings in each mode. In some aspects, the buttons **108** may be configured to activate modes based on a duration of a button press (e.g., a long press for 5 seconds). For example, after activating a desired mode, continuous pressing of the button may result in cycling between different temperature settings until the right temperature setting is selected. The first LEDs **110** may be multi-colored (e.g., bicolor, tricolor) and may indicate the currently selected temperature setting (e.g., a current temperature of the corresponding temperature mode) by displaying each temperature setting in different colors (e.g., blue, orange, red). In some aspects, the first LEDs **110** are configured to display the first temperature setting in the heating mode in a first color, a second LED **114** is configured to display the second temperature setting in the cooling mode in a second color. The first color and the second color may be different colors.

(60) In some aspects, the contrast mode may alternate between the cooling mode and heating mode. The contrast mode may include one or more cycles that alternate between the cooling mode and the heating mode for predetermined time periods. For example, a cooling mode of, e.g., 8° C. may be maintained for a certain period (e.g., 1 min) and then switched to a heating mode of, e.g., of 43° C., for a certain period (e.g., 1 min). The temperature and period settings may be a preset or user configuration settings, e.g., the temperature-controlled device **10** may be configured to communicate with a user device such as a mobile phone or computer. In some aspects, the combination of heating and cooling therapy provided by the temperature-controlled device **10** in contrast mode may be beneficial in helping a user maximize recovery of an area of their body that may be fatigued or sore from activity.

(61) When a desired mode (e.g., cooling, heating, or contrast) is selected by a user, the temperature change can be achieved quickly, e.g., it may take about 2 seconds to drop A 15° C., or it may take about 1 second to increase A 15° C. This is possible due to the above-described structural arrangement of each element forming, e.g., the air flow path from the air inlet **102** and the undercut **116** as a secondary air inlet to the air outlet **104** passing through the internal components within the housing **100**.

(62) Below the first LEDs **110**, the housing **100** further includes the charging port **112** connected to and electrically communicating with the PCB **140** to charge the one or more rechargeable batteries **130** (see FIG. 1). The second LED **114** may be located adjacent the charging port **112** and display a charging level in different colors. In some aspects, the buttons **108**, the first LEDs **110**, the charging port **112**, and the second LED **114** are on the shorter side (e.g., width direction) of the housing **100** having a generally cuboid shape, while the air outlet **104** are formed on the longer sides (e.g., length direction) of the housing **100**. In addition, one among the pair of recesses **106**, which will be described later in detail, is formed below the charging port **112** on the same side of the housing **100** without interfering with any of the control/display elements.

(63) In some aspects, although not limited, the one or more rechargeable batteries **130** may be lithium-ion batteries and may have a battery life about 60 minutes. In some aspects, the batteries **130** may be Nickel Cadmium (Ni—Cd), Nickel Metal Hydride (Ni-MH), Lithium Ion (Li-ion), Lithium Polymer (Li—Po), or other type of rechargeable batteries. In some aspects, batteries **130** may be implemented as disposable batteries. The batteries **130** may be in electric communication with the electronic components, e.g., the fan **126**, the controllable temperature element **134**, and the PCB **140**, via one or more electrical contacts. The batteries **130** may be arranged on one or more sides of the perimeter of the fan **126**, the heat sink **132**, and the controllable temperature element **134**, so as not to interfere with the air flow and thermal transfer through a central path of the internal cavity of the temperature-controlled device **10** (refer to the arrows in FIG. 19). Further and as also described above, the batteries **130** are horizontally spaced apart from each other by the fan **126**, the heat sink **132**, and the controllable temperature element **134** as well as any internal electronic components so as to form a path within housing **100** through which air may flow.

(64) For example, as shown in FIG. 7, when the temperature-controlled device **10** includes two batteries **130** supported on the support member **137** adjacent the third and fourth surfaces of the lower side cover **124**. That is, each battery **130** is seated on the corresponding wing portion **139** located at each outer side of the support member **137**. The wing portions **139** are spaced apart from each other such that internal electronic components, such as the fan **126** and the heat sink **132** can be positioned in the space between the wing portions **139** (or the two side walls **138**). As described above, such a configuration guides air flow through a path formed within the housing **100** to cool internal components in proximity to the path. This air path formed between the batteries **130** can facilitate air

to quickly connect the internal components, etc.

(65) FIGS. **9** and **10** show an internal view and an exploded view of the lower side cover **124**, respectively. The lower side cover **124** has a first opening **145**, through which the buttons **108**, the first LED **110**, the charging port **112**, and/or the second LED **114** are exposed, on a surface (e.g., third side surface) of the housing **100**. The housing **100** further includes a second opening **146** on a bottom surface facing the top surface of the temperature-controlled device **10** such that the heat spreader **118** extends out through the second opening **146**. In some aspects, the first and second side surfaces of the housing **100** extending in the length direction may be longer than the third side surface extending in the width direction. As described above, the heat spreader **118** may have a recessed portion **135** to securely receive the controllable temperature element **134**. The recessed portion **135** may reduce the weight/cost of the temperature-controlled device **10** by reducing or minimizing the use of unnecessary material.

(66) For instance, referring back to FIG. **7**, rather than having a flat surface in a solid body shape, the heat spreader **118** has the recessed portion **135** recessed on the top such that a center portion **119** is configured to protrude from the recessed portion **135** to contact the controllable temperature element **134**. The controllable temperature element **134** further includes a padded liner **136** to surround the heat spreader **118** preventing thermal energy from transferring from or to the heat spreader **118** in addition to protecting the heat spreader **118**.

(67) FIGS. **11** and **12** illustrate the portable therapeutic temperature-controlled device **10** grasped by a user's hand in various views. In particular, FIG. **12** illustrates that air flow from the air inlet **102** to the air outlet **104** is assured (e.g., the arrows).

(68) FIGS. **13** and **14** show a strap case **20** that is designed to receive the temperature-controlled device **10**. The strap case **20** is generally formed of a plastic material similar to the housing **100** and has a rectangular shape. In some aspects, the strap case **20** includes a pair of strap buckle portions **202** at opposite sides each having a buckle opening **204**, and a pair of side arms **206** curvedly extending upward. Each side arm **206** has a protrusion **208** extending toward a center opening **212** to snap-fit to the respective recess **106** of the housing **100**. In addition, the pair of strap buckle portions **202** may integrally extend away from the center opening **212** in a width direction. Accordingly, the strap case **20** can receive the temperature-controlled device **10** by engaging the protrusion **208** with the respective recess **106**. Each protrusion **208** on the pair of side arms **206** is provided with a padded layer **210** thereon to protect the friction between the protrusion **208** and the recess **106**. The padded layer **210** may be formed of a silicone material (including thermoplastic silicone, thermoset silicone, and silicone gels) or a rubber material for preventing wear and scratch between the protrusion **208** with the respective recess **106** when assembling and disassembling. In some other aspects, a plastic material may be used for the padded layer **210** using rough textures thereon.

(69) FIGS. **15-19** show an assembly of the strap case **20** and the temperature-controlled device **10** in various views. As shown in FIG. **15**, when the strap case **20** and the temperature-controlled device **10** are assembled, the air inlet **102** and the air outlet **104** are fully exposed. In addition, referring to FIG. **16** as well as FIG. **14**, the heat spreader **118** extends on a bottom side of the strap case **20** through the center opening **212** of the strap case **20** to assure the first contact to a user's body. FIG. **17** shows a top view of the assembly of the strap case **20** and the temperature-controlled device **10**, and FIG. **18** shows a bottom view of the assembly. As shown, the top cover **120** includes a plurality of openings forming the air inlet **102** allowing air to enter by the help of the fan **126**.

(70) FIG. **19** is a magnified cross-sectional view of FIG. **16** where the strap case **20** and the temperature-controlled device **10** are snap-fitted by the respective protrusion **208** and the recess **106**, with the padded layer **210** protecting the contact portion as well as securing the connection between the strap case **20** and the temperature-controlled device **10**. In addition, the battery **130** may be horizontally spaced apart from any internal electronic and metallic components of the temperature-controlled device **10**. This arrangement allows air flow (the arrows in FIG. **19**) to enter the temperature-controlled device **10** and exiting the temperature-controlled device **10** without being blocked by the battery **130**, as also described above.

(71) FIGS. **20A**, **20B**, **21A**, **21B**, **22A**, **22B**, **23A**, and **23B** illustrate a strap system for the assembly of the temperature-controlled device **10** and the strap case **20**.

(72) Referring to FIGS. **20A** and **20B**, the strap system includes a primary strap **30** having a neoprene material **302** on one side or a first side (FIG. **20A**) along the entire length and a first hook and loop material **306** (e.g., Velcro) on the entire length of the opposite side or a second side (FIG. **20B**). On the neoprene material **302**, a second hook and loop material **304** may be partially provided at one end. The primary strap **30** has a first length approximately, e.g., 100 cm (L1) and a first width approximately, e.g., 6 cm (W1), and can be fit through the buckle openings **204** of the strap buckle portions **202**. For instance, each end of the primary strap **30** can be inserted through the buckle opening **204** and folded over such that the first hook and loop material **306** can be fastened to each other, as shown in FIGS. **21A** and **21B**. When the temperature-controlled device **10**, the strap case **20**, and the primary strap **30** are assembled together, the assembly is configured to strap on a user such that heating and cooling of the controllable temperature element **134** are selectively transferred to the user's body part that is in contact with the heat spreader **118**.

(73) Additionally, with reference to FIGS. 22A and 22B, the strap system includes a secondary strap **40** having a shorter length, e.g., a second length approximately 60 cm (L2) and a second width approximately 6 cm (W2). The secondary strap **40** can be connected to the primary strap **30** to form one long strap system as shown in FIGS. 23A and 23B. Similar to the primary strap **30**, the secondary strap **40** may include a neoprene material **402** on a first side of the secondary strap **40** and a third hook and loop material **406** on the entire length of the opposite side. On the neoprene material **402**, a fourth hook and loop material **404** is partially provided at one end. The secondary strap **40** can be fit through the buckle openings **204** of the strap buckle portions **202**. For instance, each end of the secondary strap **40** can be fit through the buckle opening **204** and folded over such that the third hook and loop material **406** can be fastened to each other. The dimensions of the primary and secondary straps **30**, **40** described herein are merely exemplary values and thus not limited thereto.

(74) Alternatively, the secondary strap **40** can be coupled to the one end of the primary strap **30** by fastening the second hook and loop material **304** as a first fastening mechanism and the third hook and loop material **406** as a second fastening mechanism. That is, the adjustable strap system, using the primary strap **30** or secondary strap **40** individually, or together, allows versatile use. For instance, for use on a person with a bigger figure vs. a smaller figure, or strap around the shoulder or back vs. strap around a wrist or ankle, etc.

(75) Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” or any variant thereof, means any connection or coupling, either direct or indirect, between two or more elements; the coupling of connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description of the Aspects using the singular or plural number may also include the plural or singular number respectively. The word “or” in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

(76) The above-detailed description of aspects of the disclosure is not intended to be exhaustive or to limit the teachings to the precise form disclosed above. While specific aspects of and examples for the disclosure are described above for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. Further, any specific numbers noted herein are only examples: alternative implementations may employ differing values, measurements or ranges.

(77) Although the operations of any method(s) disclosed or described herein either explicitly or implicitly are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operations may be performed, at least in part, concurrently with other operations. In another aspect, instructions or sub-operations of distinct operations may be implemented in an intermittent and/or alternating manner.

(78) The teachings of the disclosure provided herein can be applied to other systems, not necessarily the system described above. The elements and acts of the various aspects described above can be combined to provide further aspects. Any measurements or dimensions described or used herein are merely exemplary and not a limitation on the present disclosure. Other measurements or dimensions are within the scope of the disclosure.

(79) Any patents and applications and other references noted above, including any that may be listed in accompanying filing papers, are incorporated herein by reference in their entirety. Aspects of the disclosure can be modified, if necessary, to employ the systems, functions, and concepts of the various references described above to provide yet further aspects of the disclosure.

(80) These and other changes can be made to the disclosure in light of the above Detailed Description of the Aspects. While the above description describes certain aspects of the disclosure, and describes the best mode contemplated, no matter how detailed the above appears in text, the teachings can be practiced in many ways. Details of the system may vary considerably in its implementation details, while still being encompassed by the subject matter disclosed herein. As noted above, particular terminology used when describing certain features or aspects of the disclosure should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features or aspects of the disclosure with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the disclosures to the specific aspects disclosed in the specification unless the above Detailed Description of the Aspects section explicitly defines such terms. Accordingly, the actual scope of the disclosure encompasses not only the disclosed aspects, but also all equivalent ways of practicing or implementing the disclosure under the claims.

(81) While certain aspects of the disclosure are presented below in certain claim forms, the inventors contemplate the various aspects of the disclosure in any number of claim forms. For example, while only one aspect of the disclosure is recited as a means-plus-function claim under 35 U.S.C. § 112, 9 ¶6, other aspects may likewise be

embodied as a means-plus-function claim, or in other forms, such as being embodied in a computer-readable medium. (Any claims intended to be treated under 35 U.S.C. § 112, ¶6 will include the words “means for”). Accordingly, the applicant reserves the right to add additional claims after filing the application to pursue such additional claim forms for other aspects of the disclosure.

(82) Accordingly, although exemplary aspects of the disclosure have been shown and described, it is to be understood that all the terms used herein are descriptive rather than limiting, and that many changes, modifications, and substitutions may be made by one having ordinary skill in the art without departing from the spirit and scope of the disclosure.

Claims

1. A portable temperature-controlled device comprising: a housing comprising an upper cover and a lower cover, and configured to enclose: a controllable temperature element configured to generate cooling and heating; a heat sink disposed adjacent to the controllable temperature element; a fan disposed adjacent to the heat sink and configured to direct heat away from the heat sink, wherein the fan is configured to draw outside air into the housing; a heat spreader comprising a first side and a second side, which is opposite to the first side and in contact with the controllable temperature element, wherein the first side is configured to contact a user's body part while the second side is disposed inside the housing; at least one battery, wherein the at least one battery, the heat sink, and the fan are disposed inside the housing; a temperature controller adjacent the at least one battery and connected to the controllable temperature element inside the housing; and a Printed Circuit Board (PCB) configured for electrical and data communication with the temperature controller, wherein the PCB comprises: a plurality of buttons for selectively controlling a heating mode and a cooling mode of a temperature mode by user manipulation, wherein the plurality of buttons are further configured for turning on and off the portable temperature-controlled device, wherein the heating mode includes a first temperature setting and the cooling mode includes a second temperature setting; and a first Light-Emitting Diode (LED) configured to indicate the temperature mode; wherein the housing comprises a first air inlet configured to permit air flow into the housing through the upper cover and an air outlet configured to permit the air flow to flow out of the housing through the lower cover, and wherein the first air inlet and the air outlet are in fluid communication with each other.
2. The portable temperature-controlled device of claim 1, wherein the plurality of buttons and the first LED are exposed on a surface of the housing.
3. The portable temperature-controlled device of claim 2, wherein the first temperature setting includes a temperature range between 35° C. and 43° C., and a second temperature setting includes a temperature range between 8° C. and 16° C., and wherein the first LED includes a plurality of LEDs configured to display different colors from each other, and the plurality of LEDs are configured to indicate a current temperature of the temperature mode.
4. The portable temperature-controlled device of claim 2, wherein the first air inlet is defined on a top surface of the portable temperature-controlled device, wherein the first air inlet is configured to allow the air flow into the housing, wherein the air outlet is defined on at least one of a first or second side surface of the portable temperature-controlled device, wherein the air outlet is configured to allow the air flow to exit the housing, and wherein the housing further comprises a first opening on a third side surface and a second opening on a bottom surface facing the top surface of the portable temperature-controlled device.
5. The portable temperature-controlled device of claim 4, wherein the PCB further includes a charging port disposed below the first LED and exposed outside the housing through the first opening of the housing, the charging port electrically communicating with the PCB, wherein the heat spreader is configured to extend out through the second opening, wherein the PCB includes a second LED configured to indicate a charging level in different colors, and wherein the second LED is disposed adjacent the charging port.
6. The portable temperature-controlled device of claim 1, wherein the housing further comprises a support member, wherein the at least one battery is horizontally spaced apart from the fan and the heat sink and supported on the support member, wherein the at least one battery is positioned so as to permit air flow between the fan and the heat spreader, wherein the support member has an opening in a center, the opening being smaller than the heat sink but larger than the controllable temperature element, and wherein the heat spreader is connected to a bottom side of the support member.
7. The portable temperature-controlled device of claim 1, wherein the housing further comprises a middle cover, wherein the upper cover has a convex shape and is configured to be assembled to the middle cover having an undercut, wherein the undercut is positioned along an upper edge of the middle cover adjacent the upper cover, and wherein the undercut further defines a second air inlet configured to permit additional air flow to enter the housing between the upper cover and the middle cover, and wherein the middle cover and the lower cover are assembled forming a partially enclosed space of the housing together with the upper cover.

8. The portable temperature-controlled device of claim 7, wherein the upper cover comprises a first plurality of vents, wherein the fan is configured to cause the air flow to flow into the housing through the first plurality of vents, and wherein the lower cover includes: a second plurality of vents configured to permit the air flow to exit out from the housing, wherein the second plurality of vents are defined on first and second surfaces of the lower cover arranged longitudinally, and wherein the first and second surfaces are positioned facing each other; and a pair of recesses defined on third and fourth surfaces, wherein the third and fourth surfaces are positioned facing each other in a width direction.

9. The portable temperature-controlled device of claim 8, wherein the at least one battery is spaced apart from the fan, the heat sink, and the controllable temperature element, wherein the at least one battery includes two batteries adjacent to the third and fourth surfaces, and wherein the first plurality of vents, the fan, the heat sink, and the second plurality of vents are arranged to define an air flow passage for the air flow from the first plurality of vents to the second plurality of vents through the fan and the heat sink.

10. The portable temperature-controlled device of claim 1, wherein the second side of the heat spreader is disposed inside the housing and includes a padded liner that surrounds the controllable temperature element, and wherein the heat spreader is configured to receive thermal energy from the controllable temperature element.

11. A portable temperature-controlled device comprising: a housing comprising an upper cover, a lower cover, and a support member, wherein the housing is configured to enclose: a controllable temperature element configured to generate cooling and heating; a heat sink disposed adjacent to the controllable temperature element; a fan disposed adjacent to the heat sink and configured to direct heat away from the heat sink, wherein the fan is configured to draw outside air into the housing; a heat spreader connected to a bottom side of the support member, the heat spreader comprising a first side and a second side, which is opposite to the first side and in contact with the controllable temperature element, wherein the first side is configured to contact a user's body part while the second side is disposed inside the housing; at least one battery supported on the support member and horizontally spaced apart from the fan and the heat sink, wherein the at least one battery, the heat sink, and the fan are disposed inside the housing, and wherein the at least one battery is positioned so as to permit air flow between the fan and the heat spreader; and a temperature controller adjacent the at least one battery and connected to the controllable temperature element inside the housing, wherein the housing comprises a first air inlet configured to permit air flow into the housing through the upper cover and an air outlet configured to permit the air flow to flow out of the housing through the lower cover, wherein the first air inlet and the air outlet are in fluid communication with each other, and wherein the support member has an opening in a center, the opening being smaller than the heat sink but larger than the controllable temperature element.

12. A portable temperature-controlled device comprising: a housing comprising: an upper cover having a convex shape; a middle cover having an undercut positioned along an upper edge of the middle cover adjacent the upper cover; a lower cover, wherein the middle cover and the lower cover are assembled to form a partially enclosed space of the housing together with the upper cover; a first air inlet configured to permit air flow into the housing through the upper cover; and an air outlet in fluid communication with the first air inlet, the air outlet configured to permit the air flow to flow out of the housing through the lower cover, wherein the housing is configured to enclose: a controllable temperature element configured to generate cooling and heating; a heat sink disposed adjacent to the controllable temperature element; a fan disposed adjacent to the heat sink and configured to direct heat away from the heat sink, wherein the fan is configured to draw outside air into the housing; a heat spreader comprising a first side and a second side, which is opposite to the first side and in contact with the controllable temperature element, wherein the first side is configured to contact a user's body part while the second side is disposed inside the housing; at least one battery, wherein the at least one battery, the heat sink, and the fan are disposed inside the housing; and a temperature controller adjacent the at least one battery and connected to the controllable temperature element inside the housing.

13. The portable temperature-controlled device of claim 12, wherein the undercut further defines a second air inlet configured to permit additional air flow to enter the housing between the upper cover and the middle cover.

14. The portable temperature-controlled device of claim 13, wherein the fan is configured to cause the air to flow into the housing through a first plurality of vents, and wherein the lower cover includes: first and second surfaces positioned facing each other in a longitudinal direction; a second plurality of vents defined on the first and second surfaces and configured to permit the air flow to exit from the housing; third and fourth surfaces positioned facing each other in a width direction; and a pair of recesses defined on the third and fourth surfaces.

15. The portable temperature-controlled device of claim 14, wherein the at least one battery is spaced apart from the fan, the heat sink, and the controllable temperature element, wherein the at least one battery includes two batteries adjacent to the third and fourth surfaces, and wherein the first plurality of vents, the fan, the heat sink, and the second plurality of vents are arranged to define an air flow passage for the air flow from the first plurality of vents to the second plurality of vents through the fan and the heat sink.

16. A portable temperature-controlled device comprising: a housing comprising an upper cover and a lower cover,

and configured to enclose: a controllable temperature element configured to generate cooling and heating; a heat sink disposed adjacent to the controllable temperature element; a fan disposed adjacent to the heat sink and configured to direct heat away from the heat sink, wherein the fan is configured to draw outside air into the housing; a heat spreader configured to receive thermal energy from the controllable temperature element, the heat spreader comprising a first side and a second side, which is opposite to the first side and in contact with the controllable temperature element, wherein the first side is configured to contact a user's body part while the second side is disposed inside the housing, wherein the heat spreader includes a padded liner that surrounds the controllable temperature element; at least one battery, wherein the at least one battery, the heat sink, and the fan are disposed inside the housing; and a temperature controller adjacent the at least one battery and connected to the controllable temperature element inside the housing, wherein the housing comprises a first air inlet configured to permit air flow into the housing through the upper cover and an air outlet configured to permit the air flow to flow out of the housing through the lower cover, and wherein the first air inlet and the air outlet are in fluid communication with each other.
